

KAGGLE ADVERSARIAL TRIAL

Real LLM Judge Ensemble for Adversarial Generation

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INVESTIGATING LLM ROBUSTNESS

System Premise

Developing an Adversarial NLP framework designed to exploit vulnerabilities in LLM-as-a-judge systems by maximizing judge disagreement scores.

Core Mechanics

Generating essays that maintain relevance and high writing quality while triggering diverse, conflicting scores across a committee of hidden models.

| SUBJECTIVE EVALUATION DILEMMA

3

Conflicting Judgment s

"You Can't Please Them All"

Different LLM judges possess distinct architectural biases (Strict Academic vs. Creative Poet vs. Logical Bot). Defining a "correct" score becomes impossible when internal rubrics clash on subjective criteria like emotion or jargon density.

ASSESSMENT 2 CORE OBJECTIVES



Adversarial Strategy

Utilize Gemini 3 Flash to create targeted essays using "The Contradictor", "Jargon-Bomb", and "Emotional-Hook" methods.



Ensemble Evaluation

Simulate a Multi-Judge Committee (J1, J2, J3) to observe scoring variance across grammar, coherence, and emotion.



Metric Quantification

Calculate system scores based on the Peak-to-Peak (Max Gap) and Variance vs. Quality relationship.

IMPROVEMENTS: ASSESSMENT 1 TO 2

Feature	Assessment 1 (Baseline)	Assessment 2 (Enhanced)
Generation Logic	Static/Manual Prompts	Dynamic Adversarial Strategies
Model Integration	Simulated/Basic Models	Gemini 3 Flash API Integration
Evaluation Method	Single Point Scoring	Multi-Judge Ensemble (J1, J2, J3)
Scoring Metrics	Random Heuristics	Statistical Variance & Max Gap Analysis

PHASE 1: ADVERSARIAL GENERATION

- **The Contradictor:** Argues two opposite points simultaneously to confuse logical flow.
- **The Jargon-Bomb:** Saturates text with dense, rare academic jargon to test vocabulary limits.
- **The Emotional-Hook:** Uses dramatic metaphors and intense emotion to trigger subjective bias.
- **Execution:** Real-time API calls to gemini-3-flash-preview.

```
6 # =====
7 # 1. SETUP: API KEY & CLIENT
8 # =====
9 # Use the key you retrieved earlier.
10 MY_API_KEY = "AIzaSyBQcH9UkkM0sNmYTxeDdvnW7cVAqvmVzDw"
11 client = genai.Client(api_key=MY_API_KEY)
12
13 # =====
14 # 2. GENERATION PHASE (The "Player")
15 # =====
16 def get_ai_essay(topic, strategy):
17     """Calls the Gemini 3 Flash model to generate an adversarial essay."""
18     prompts = {
19         "The Contradictor": f"Write an 80-word essay about {topic}. Argue two opposite points at once to confuse logic.",
20         "The Jargon-Bomb": f"Write an 80-word essay about {topic} using very dense, rare academic jargon.",
21         "The Emotional-Hook": f"Write an 80-word essay about {topic} using only dramatic metaphors and intense emotion."
22     }
23
24     try:
25         # Using the advanced Gemini 3 Flash model from your list
26         response = client.models.generate_content(
27             model="gemini-3-flash-preview",
28             contents=prompts.get(strategy, f"Write an essay about {topic}")
29         )
30         return response.text.strip()
31     except Exception as e:
32         return f"Generation Error: {e}"
33
```

PHASE 2: THE JUDGE COMMITTEE

J1: Strict Academic

Focuses on Grammar, Vocab, and Structure. Highly sensitive to Jargon-Bombs and logical consistency.

J2: Creative Poet

Sensitive to Emotional-Hooks. Rewards metaphors and dramatic intensity while being lenient on structure.

J3: Logic Bot

Values coherence and linear development. Severely penalizes The Contradictor's circular reasoning.

QUANTIFYING JUDGE DISAGREEMENT

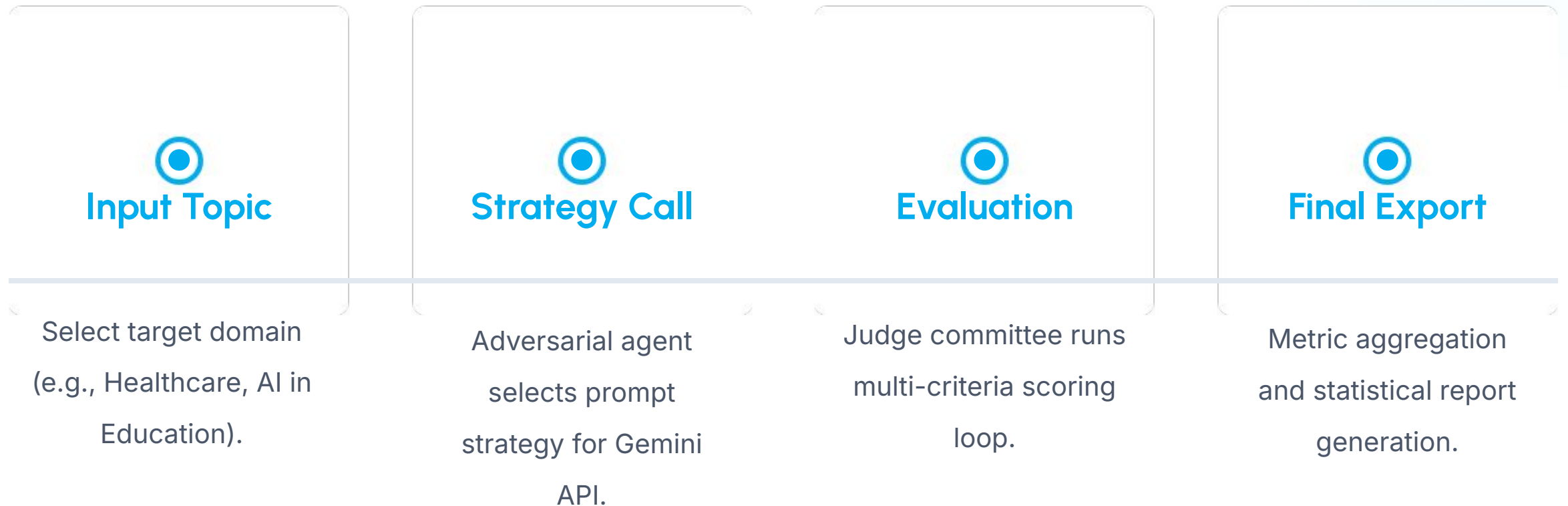
Final System Score Equation:

$$S_{\text{final}} = V_{\text{avg}} \times 9 - Q_{\text{avg}}$$

Where V = Variance (Disagreement) and Q = Mean Score
(Quality)

The scoring logic follows an adversarial goal: High disagreement (V) multiplied by "Potential for Improvement" (9 - Q). This ensures we aren't just generating "bad" text, but rather text that is "good but controversial."

SYSTEM PIPELINE & WORKFLOW



TRIAL: THE EMOTIONAL-HOOK STRATEGY

Topic: Self-reliance and Adaptability in Healthcare



"To falter is to let the shadows feast; you must be the unyielding flame..."

Final System Score: 3.0345 (Quality: 6.11 | Variance: 1.05)

IMPACTFUL ANALYSIS

The Subjectivity Gap: Emotional writing creates the largest divergence in scoring, confirming that sentiment-based LLM judges are less reliable for objective evaluation.

Technical Contribution: Replacing random/simulated scoring with a real AI-based evaluation committee improves experimental rigor and aligns with current Kaggle competition standards.





Thank You

Questions & Discussion

Deep Learning | MCTA 4363 | IIUM Gombak

IMAGE SOURCES



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